PROJECT REPORT

RetailPulse

AI-Powered Customer Analytics & Demand Forecasting Platform

An end-to-end retail intelligence system — customer segmentation, demand forecasting, churn prediction, and inventory optimization, delivered through an interactive dashboard backed by a full MLOps pipeline.

779K	5,878	₹17.4M	6
TRANSACTIONS	CUSTOMERS	REVENUE ANALYZED	ML MODELS

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Data Science & Analytics Intern · Zidio Development · June 2026

retail-pulse-ai.streamlit.app

1. Project Overview

Vision & objectives. RetailPulse is an AI-powered retail analytics platform that ingests a year of raw transaction data and answers four core business questions through a single interactive dashboard backed by a production-grade MLOps pipeline:

Business question	How RetailPulse answers it
Who are my customers?	RFM scoring + K-Means / DBSCAN segmentation
What will I sell next?	Prophet + LSTM hybrid demand forecasting
Who is about to leave?	XGBoost churn model tuned with Optuna
What should I restock?	EOQ, safety-stock & reorder-point optimization

It additionally estimates **customer lifetime value** (BG/NBD + Gamma-Gamma), generates **natural-language insights** (LLM-powered), and runs an **automated retraining pipeline** that retrains models when data drift is detected.

Target users / use cases.

- **Executives** — a unified command center with health scorecards, revenue-at-risk, and recommended actions.
- **Analysts** — segmentation, churn, and forecasting views for campaign and demand planning.
- **Data scientists** — model diagnostics, SHAP explainability, and MLflow experiment tracking.
- **Operations** — inventory reorder queue and threshold-based stockout / churn / revenue alerts.

Business value delivered. Built on a real dataset of **779,426 transactions**, **5,878 customers**, and **36,969 orders** (₹17.37M total revenue), the platform surfaces **₹8.08M of projected 12-month customer lifetime value**, flags customers at churn risk with high recall, and identifies the units needing reorder across the catalog — converting raw history into prioritized, revenue-protecting actions.

Non-functional goals.

- **Accuracy gate** — releases blocked if churn AUC or forecast error regress below `models/baseline_metrics.json`.
- **Responsiveness** — three-layer data cascade (Redis cache → PostgreSQL → CSV fallback) keeps the dashboard fast and degrades gracefully if a backing service is down.
- **Scalability** — containerized stack with Kubernetes manifests for horizontal scaling.

- **Security** — JWT authentication with four roles; every page enforces login and role permission; no secrets committed.

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2. Key Features

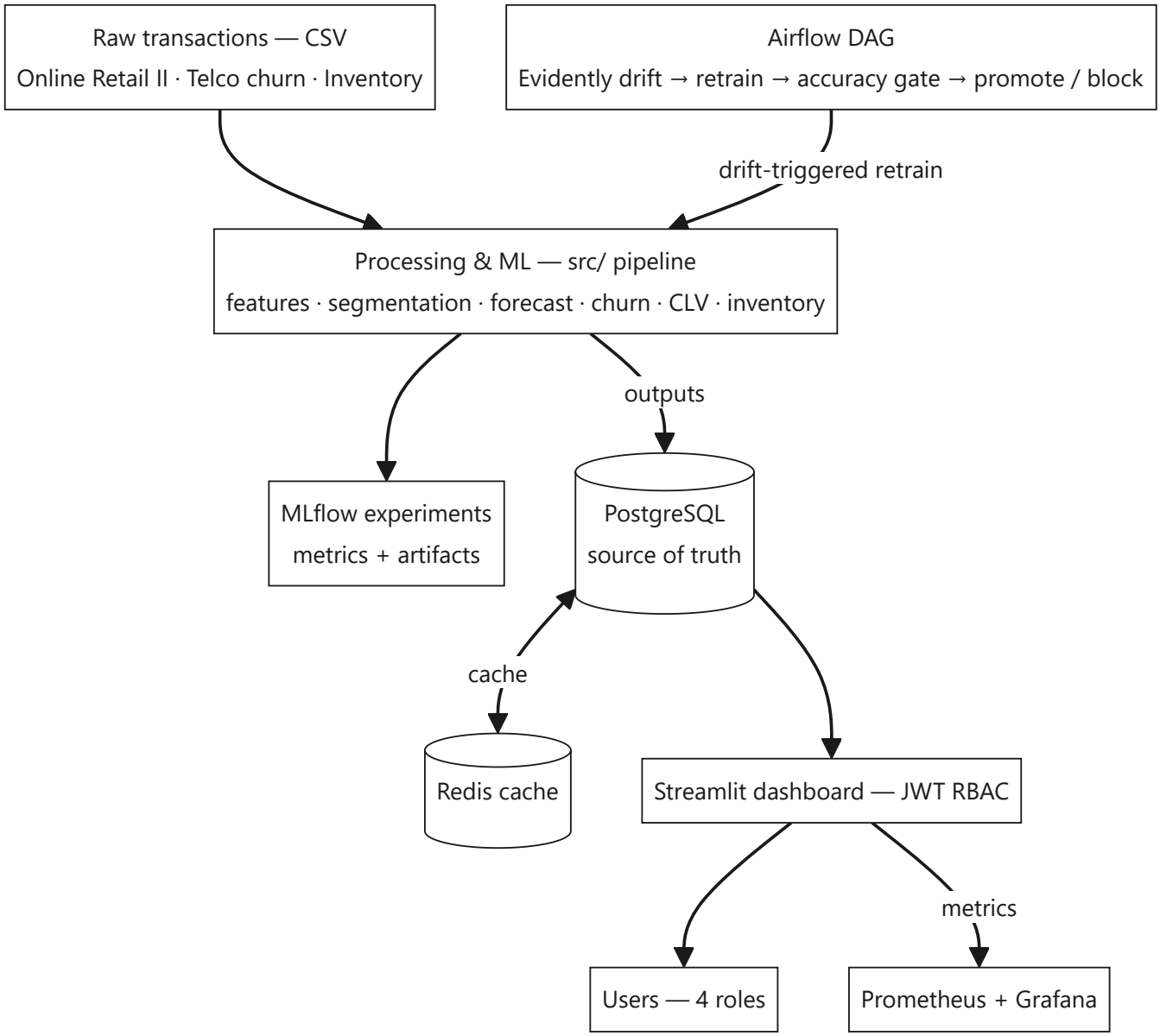
ID	Feature	Description	Acceptance Criteria
F1	Customer segmentation	RFM scoring with K-Means + DBSCAN, profiled into business labels (Champions, Loyal, Churned, Outlier)	Every customer assigned a cluster + business label; clusters reproducible from `run_segmentation`
F2	Demand forecasting	Prophet + LSTM residual hybrid forecasts daily revenue	Hybrid beats baseline on validation; MAPE / MAE / RMSE logged to MLflow
F3	Churn prediction	Optuna-tuned XGBoost with SHAP explainability and per-customer probabilities	Test AUC ≥ baseline (0.928); SHAP ranking produced for every model
F4	Inventory optimization	EOQ, safety stock & reorder points from forecasted demand	Each SKU classified (optimal / stockout-risk / overstock) with units-to-order
F5	Customer lifetime value	BG/NBD + Gamma-Gamma 12-month CLV with probability-alive	CLV + P(alive) produced for every active customer
F6	Interactive dashboard	Multipage Streamlit app — overview, segmentation, churn, forecasting, inventory, CLV, alerts, command center	All pages render with DB or CSV fallback; no hard dependency on a live service
F7	Role-based access control	JWT auth, four roles, per-page permission enforcement	Unauthorized role blocked from restricted pages; session persists across refresh
F8	AI insights & alerts	LLM-generated briefings, NL Q&A, threshold email alerts (churn / stockout / revenue)	Alerts fire on threshold breach; sends logged to history
F9	MLOps loop	MLflow tracking, Evidently drift detection, Airflow drift-triggered retrain behind an accuracy gate	Drift report generated; retrain only promotes models that clear the gate
F10	CI/CD & deployment	Lint + test + accuracy gate → image build → Kubernetes deploy; hosted on Streamlit Cloud	Accuracy regression blocks release; green pipeline deploys automatically

3. Technology Stack

Category	Technology	Rationale / Alternatives
Forecasting	**Prophet + PyTorch Lightning (LSTM)**	Prophet captures trend/seasonality interpretably; LSTM models the residual non-linearity. *Alt: ARIMA (weaker on multi-seasonality), pure LSTM (less interpretable).*
Churn	**XGBoost + Optuna**	Strong tabular performance, native feature importance; Optuna for efficient hyperparameter search. *Alt: LogisticRegression (lower AUC), RandomForest.*
Segmentation	**scikit-learn (K-Means, DBSCAN)**	K-Means for clean business segments, DBSCAN to catch outliers. *Alt: hierarchical clustering (costlier at scale).*
Lifetime value	**lifetimes (BG/NBD + Gamma-Gamma)**	Established probabilistic CLV models for non-contractual retail. *Alt: heuristic historical CLV (no forward projection).*
Explainability	**SHAP**	Per-customer, theory-grounded attributions for the churn model.
Experiment tracking	**MLflow**	Metrics, params & artifacts per run; backs the release-gate baseline.
Drift detection	**Evidently AI**	Ready-made data/target drift reports feeding the retrain trigger.
Orchestration	**Apache Airflow**	DAG schedules drift checks and gated retraining.
Dashboard	**Streamlit + Plotly**	Fast multipage analytics UI in pure Python; Plotly for interactivity.
Auth	**JWT**	Stateless role-based access control across pages.
Data stores	**PostgreSQL + Redis**	Postgres as source of truth; Redis cache for low-latency reads.
Infra	**Docker Compose, Kubernetes**	Reproducible local stack and scalable cluster deploy.
Monitoring	**Prometheus + Grafana**	Metrics scraping and dashboards.
CI/CD	**GitHub Actions, Ruff, pytest**	Lint + test on every push; accuracy-gated deploy.
Hosting	**Streamlit Cloud + Supabase + Upstash**	Managed demo: Streamlit Cloud app, Supabase Postgres, Upstash Redis (TLS).

4. Architecture Overview

High-level data flow and MLOps pipeline:



Data flow. Loaders read through a three-layer cascade — **Redis cache** → **PostgreSQL (source of truth)** → **CSV fallback** — so the app stays responsive and degrades gracefully if a backing service is unavailable. The ML pipeline (`src/`) writes processed outputs and logs every run to MLflow; the Airflow DAG retrains behind an accuracy gate when Evidently detects drift.

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5. Detailed Execution Timeline

A 4-week internship build, executed day-by-day:

Week 1 — Foundation & baseline models

Day	Deliverable
1–2	EDA, data cleaning, feature engineering, validation utilities + README
3	Customer segmentation — K-Means & DBSCAN
4	Time-series forecasting prep pipeline
5	Baseline Prophet demand forecast
6	LSTM forecaster (PyTorch Lightning)
7	**Checkpoint:** EDA report, cleaned dataset, baseline models logged to MLflow

Week 2 — Advanced models & MLOps core

Day	Deliverable
8	Hybrid forecasting (Prophet + LSTM ensemble)
9	Churn model (XGBoost + SHAP explainability)
10	Inventory optimization from forecasted demand
11	Feature-importance analysis + Optuna tuning
12	Drift detection (Evidently AI)
13	Automated retraining pipeline (Airflow)
14	**Checkpoint:** forecasting & churn models ready, optimization logic implemented

Week 3 — Dashboard

Day	Deliverable
15	Streamlit skeleton — multipage layout
16	Forecasting visualizations + what-if analysis
17	Segmentation & churn-risk dashboard
18	Inventory recommendations UI
19	Real-time metrics & alerts
20	Export functionality (CSV / PDF) — **Week 3 complete**

Week 4 — Production, infra & deployment

Day	Deliverable
21–22	Role-based access (4 roles), PostgreSQL integration, project hardening
23	CI/CD pipeline, Kubernetes manifests, Docker multi-stage builds
24–25	Streamlit Cloud deployment + Supabase migration; refresh/session & data-fallback fixes
26	Prometheus + Grafana monitoring
27	Load testing, final accuracy validation, accuracy gate as a CD release gate

Milestones: Week-1 baseline models ✓ · Week-2 MLOps loop ✓ · Week-3 full dashboard ✓ · Week-4 production deploy + monitoring ✓.

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6. Technical Highlights

Model performance (real values, from MLflow)

Churn — Optuna-tuned XGBoost

Metric	Value
Test AUC-ROC	**0.927**
Cross-val AUC	0.945
F1	0.816
Recall	0.970
Precision	0.704
Precision @ top-20%	0.766

High recall (0.97) is intentional — the cost of missing a churner outweighs a false alarm, so the model is tuned to catch nearly all at-risk customers.

Demand forecast — Prophet + LSTM hybrid

Metric	Value
MAPE	31.2%
MAE	₹10,922
RMSE	₹18,981

Customer lifetime value (BG/NBD + Gamma-Gamma)

Metric	Value
Mean 12-month CLV	₹1,930
Median 12-month CLV	₹872
Total projected CLV	₹8.08M
High-value customers	33%
Mean P(alive)	0.82

Inventory optimization

Metric	Value
SKUs at optimal stock	38.75%
SKUs at stockout risk	61.25%
Total units to order	931,134

Segmentation (5,878 customers). K-Means produced three actionable segments plus outliers: **Loyal Customers (2,509)**, **Churned (1,862)**, **Champions (1,469)**, **Outliers (38)**; DBSCAN isolated the same 38 outliers from a dense core.

Feature engineering

RFM features (recency, frequency, monetary) plus behavioral signals — `purchase_regularity`, `n_months_active`, `tenure`, `avg_days_between_orders`, `spend_trend_90d`, `recent_freq_90d`, and `revenue_last_30d`. A 5-method importance consensus (XGB gain, XGB weight, SHAP, permutation, mutual information) ranked **monetary, purchase_regularity, and frequency** as the top churn drivers.

MLOps practices

- **MLflow** — every run logs metrics, params, and artifacts across experiments spanning forecasting, churn, segmentation, CLV, inventory, and drift detection.
- **Evidently drift detection** — latest report flagged **66.7% revenue feature drift** and 23% churn feature drift, no target drift — the signal that triggers retraining.
- **Airflow retrain pipeline** — drift-triggered retrain promotes a new model **only if it clears the accuracy gate** against `baseline_metrics.json`.

Challenges faced & how solved

- **Forecast accuracy** — a single Prophet model underfit non-linear residuals; solved with a **Prophet + LSTM hybrid** that models the residual.
- **Class imbalance in churn** — tuned with Optuna toward recall so few churners are missed.
- **Service availability** — built a **Redis → PostgreSQL → CSV cascade** so the dashboard never hard-fails when a backend is down.
- **Streamlit Cloud session loss on refresh** — fixed cookie-based session persistence so login survives navigation and refresh.
- **CI reliability** — pinned CPU-only torch and auto-skip data-dependent tests so CI runs without the 94 MB raw dataset.

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7. Deployment & Operations

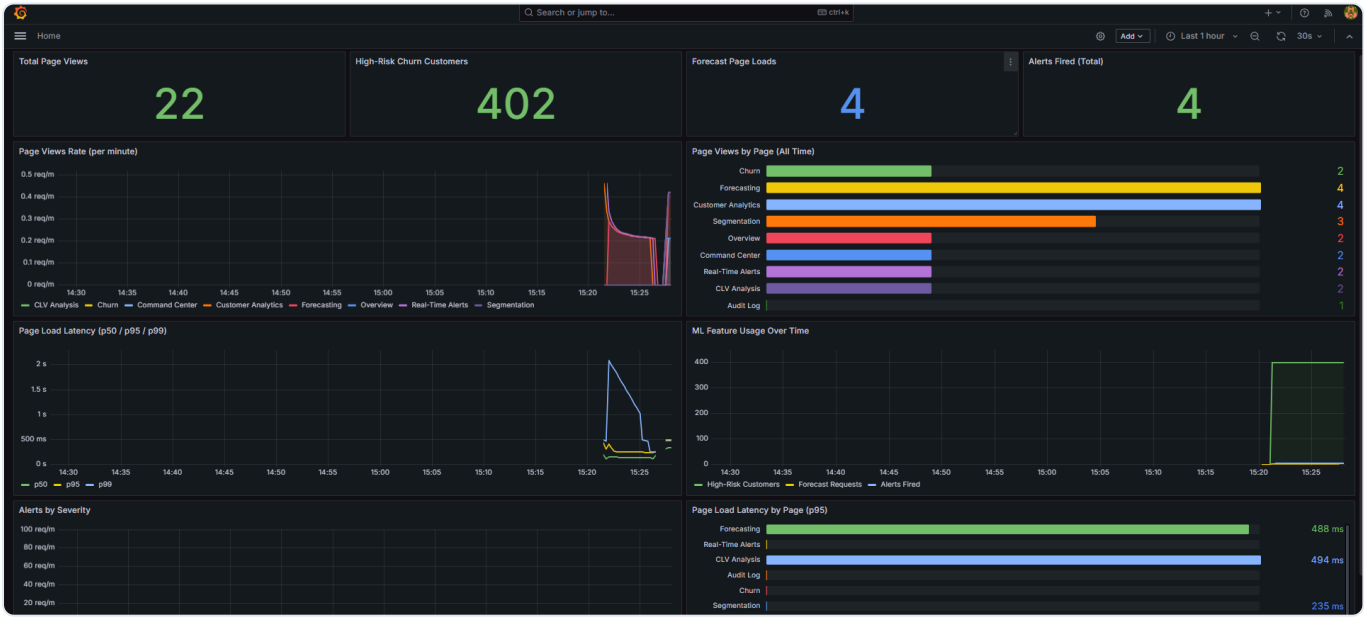
Platform. The hosted demo runs on **Streamlit Community Cloud** with managed backing services — **Supabase** (PostgreSQL, connection pooler) and **Upstash** (Redis, TLS). Secrets (`DATABASE_URL`, `REDIS_URL`, `GROQ_API_KEY`, `ALERT_EMAIL_*`) are injected through the platform's secrets manager and never committed. For self-hosting, the `k8s/` manifests cover the same stack on a cluster.

CI/CD pipeline.

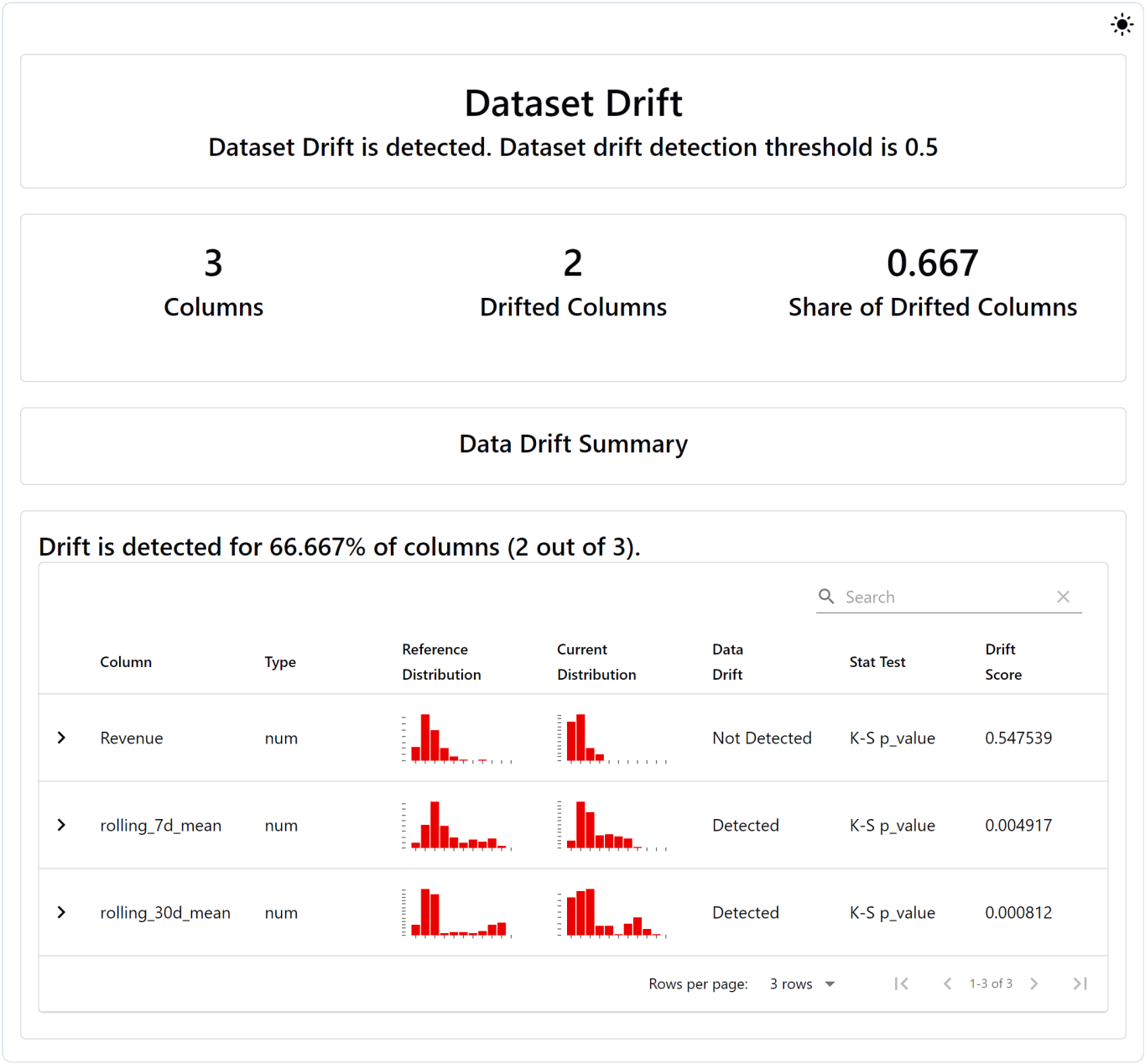
- **CI (`ci.yml`)** — Ruff lint + pytest on every push and PR, with concurrency cancellation and pip caching.
- **CD (`cd.yml`)** — staged release:
- **Accuracy gate** (`run_validation`) — a regression below `baseline_metrics.json` blocks everything downstream.
- **Build & push** — multi-stage app image + MLflow image to GitHub Container Registry (ghcr.io), with Actions-cache layer reuse.
- **Deploy to Kubernetes** — gated behind a `production` environment (optional manual approval).

Monitoring / drift detection. Prometheus scrapes service metrics into Grafana dashboards; Evidently drift reports (in `reports/drift/`) feed the Airflow retrain trigger, closing the MLOps loop.

Grafana monitoring dashboard — live Prometheus metrics: page views, forecast loads, alert counts, and page-load latency (p50 / p95 / p99) broken down per page.



Evidently drift report — revenue drift detected across 2 of 3 monitored columns (66.7% share), the signal that triggers a gated retrain.



Dataset Drift

Dataset Drift is detected. Dataset drift detection threshold is 0.5

3	2	0.667
Columns	Drifted Columns	Share of Drifted Columns

Data Drift Summary

Drift is detected for 66.667% of columns (2 out of 3).

Search

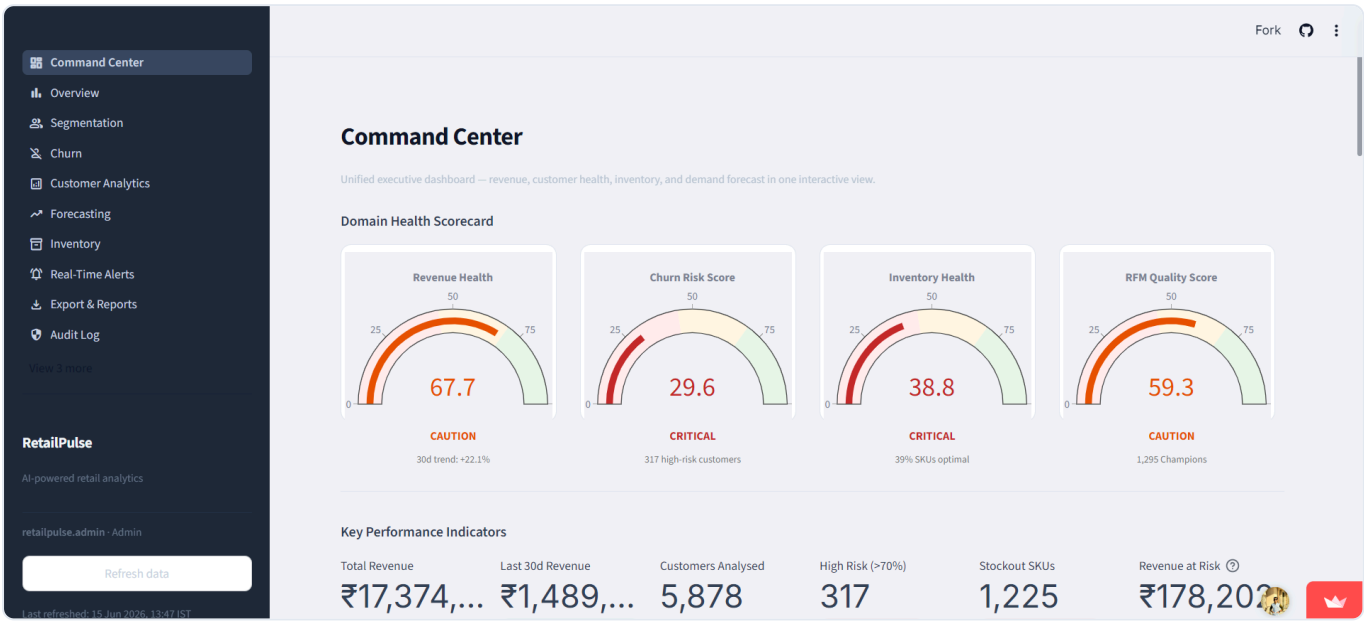
Column	Type	Reference Distribution	Current Distribution	Data Drift	Stat Test	Drift Score
Revenue	num			Not Detected	K-S p_value	0.547539
rolling_7d_mean	num			Detected	K-S p_value	0.004917
rolling_30d_mean	num			Detected	K-S p_value	0.000812

Rows per page: 3 rows 1-3 of 3

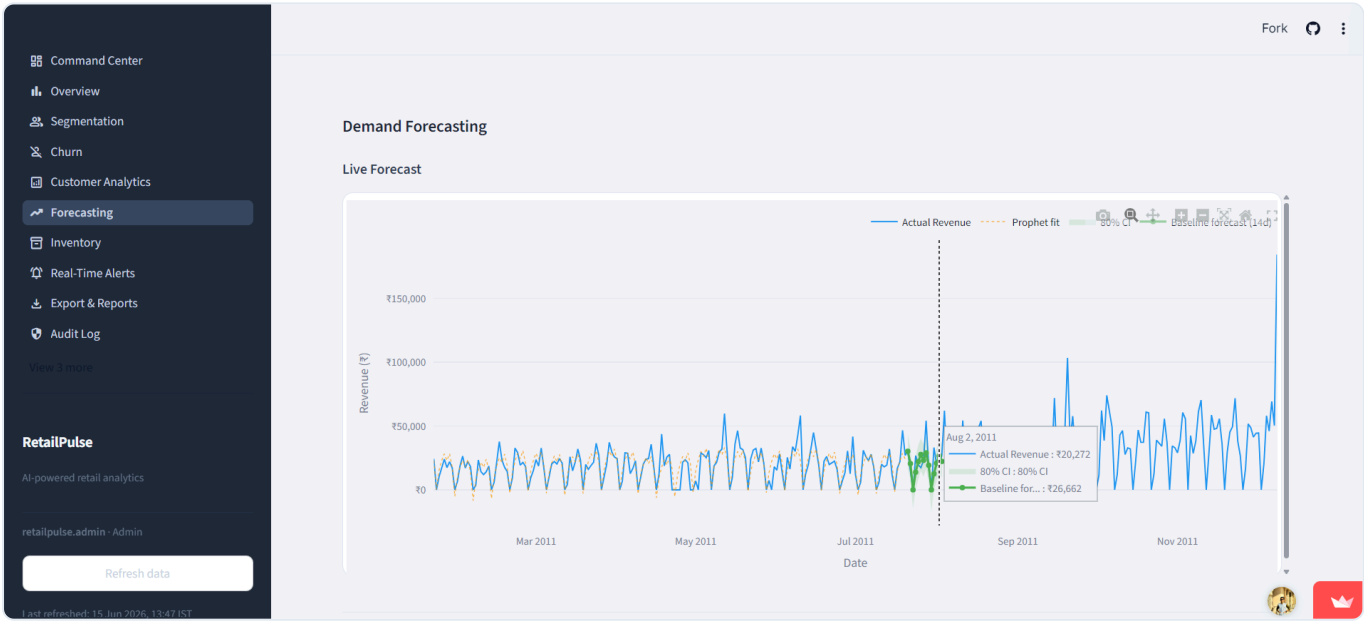
8. Visuals

Live dashboard

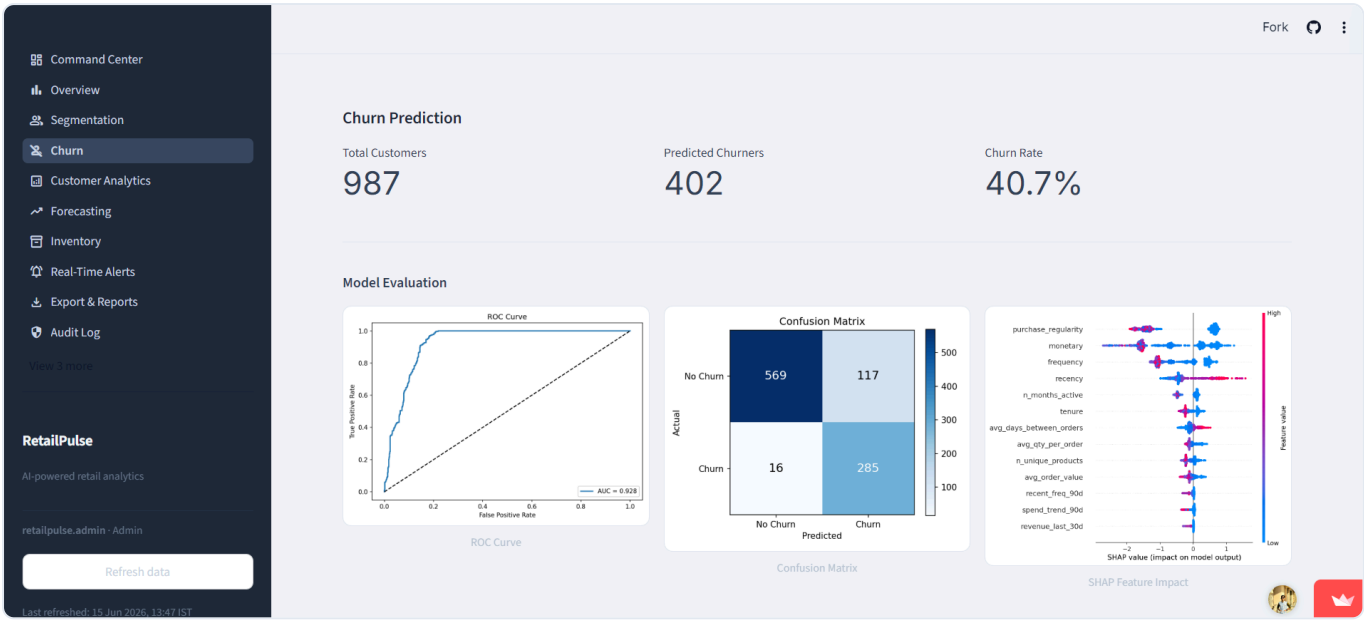
Command Center — unified executive view with health scorecards, revenue-at-risk, and a recommendation engine.



Forecasting — Prophet + LSTM hybrid demand forecast with what-if analysis.



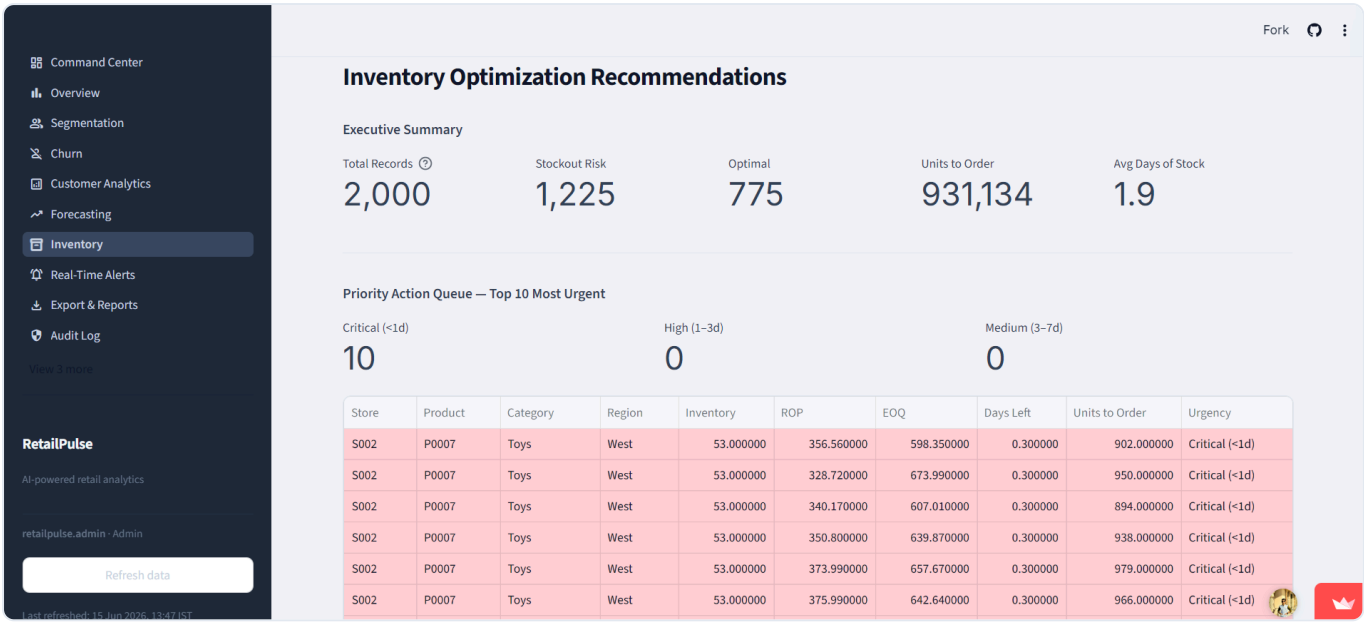
Churn — per-customer churn probabilities from the Optuna-tuned XGBoost model.



Segmentation — RFM scoring with K-Means / DBSCAN clusters, profiled by business value.

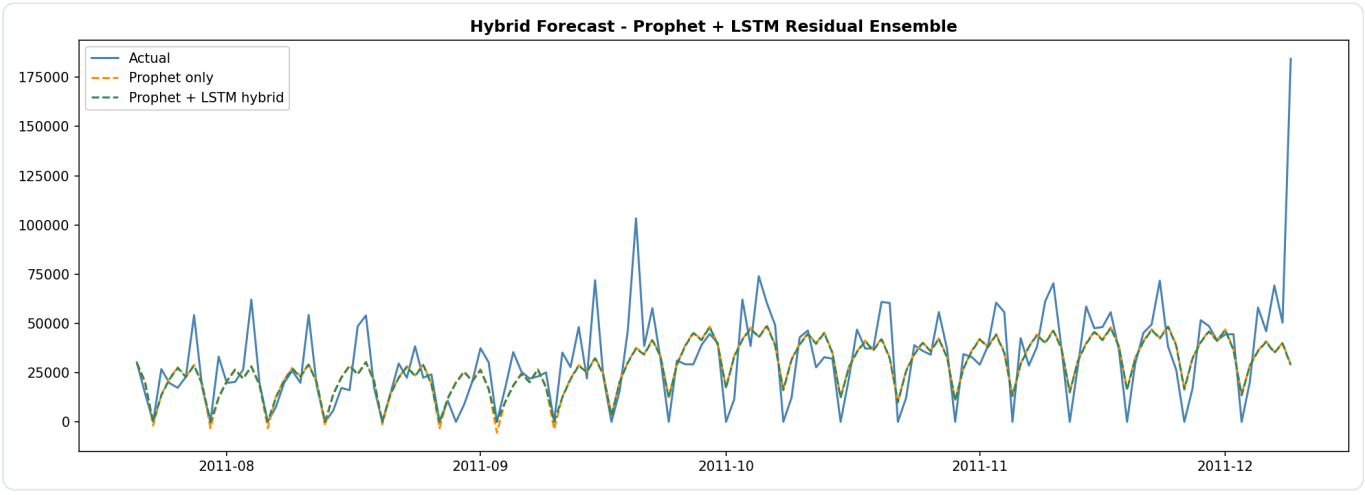


Inventory — reorder points, safety stock, and an urgency-tiered restock queue.

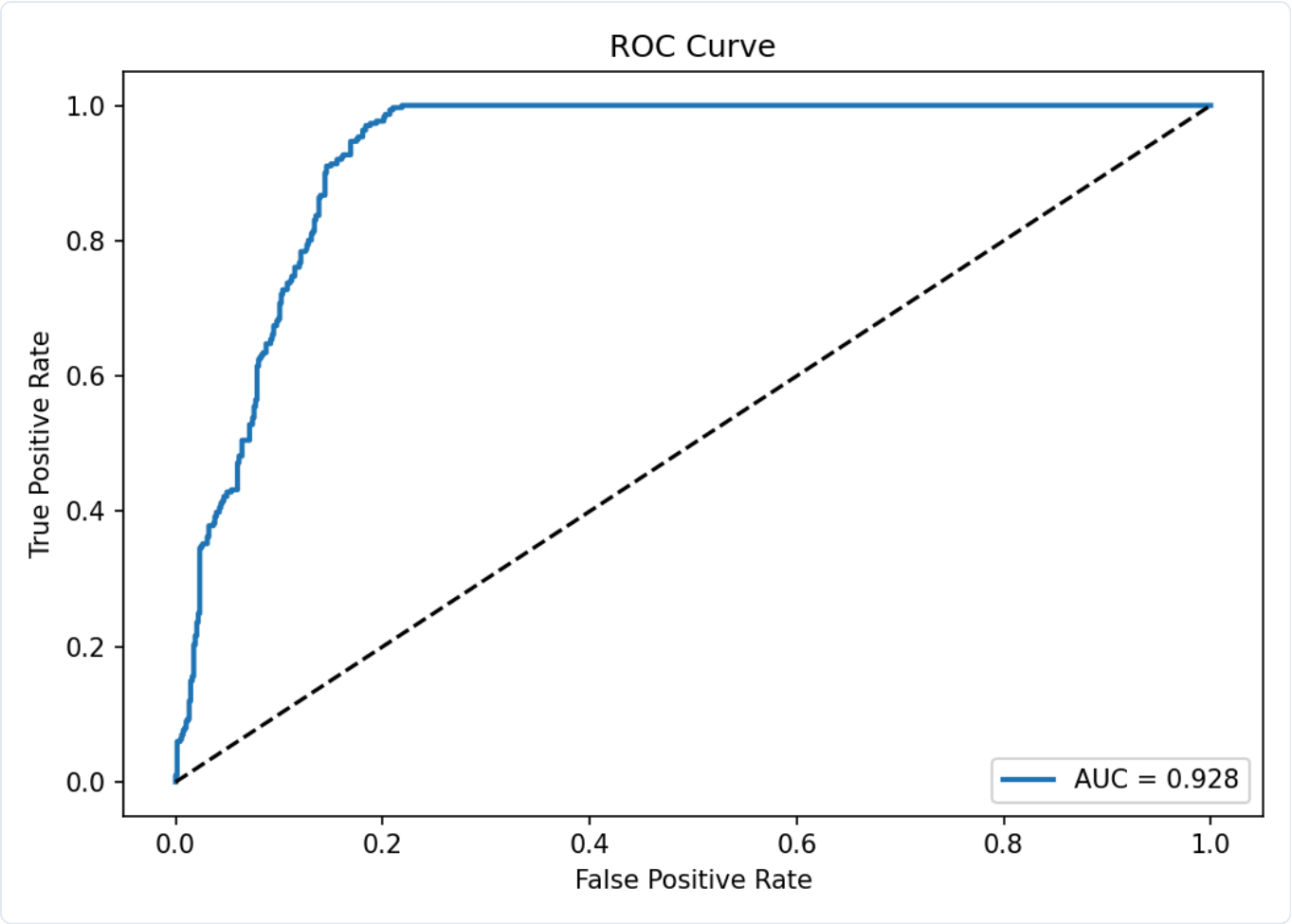


Model outputs & EDA

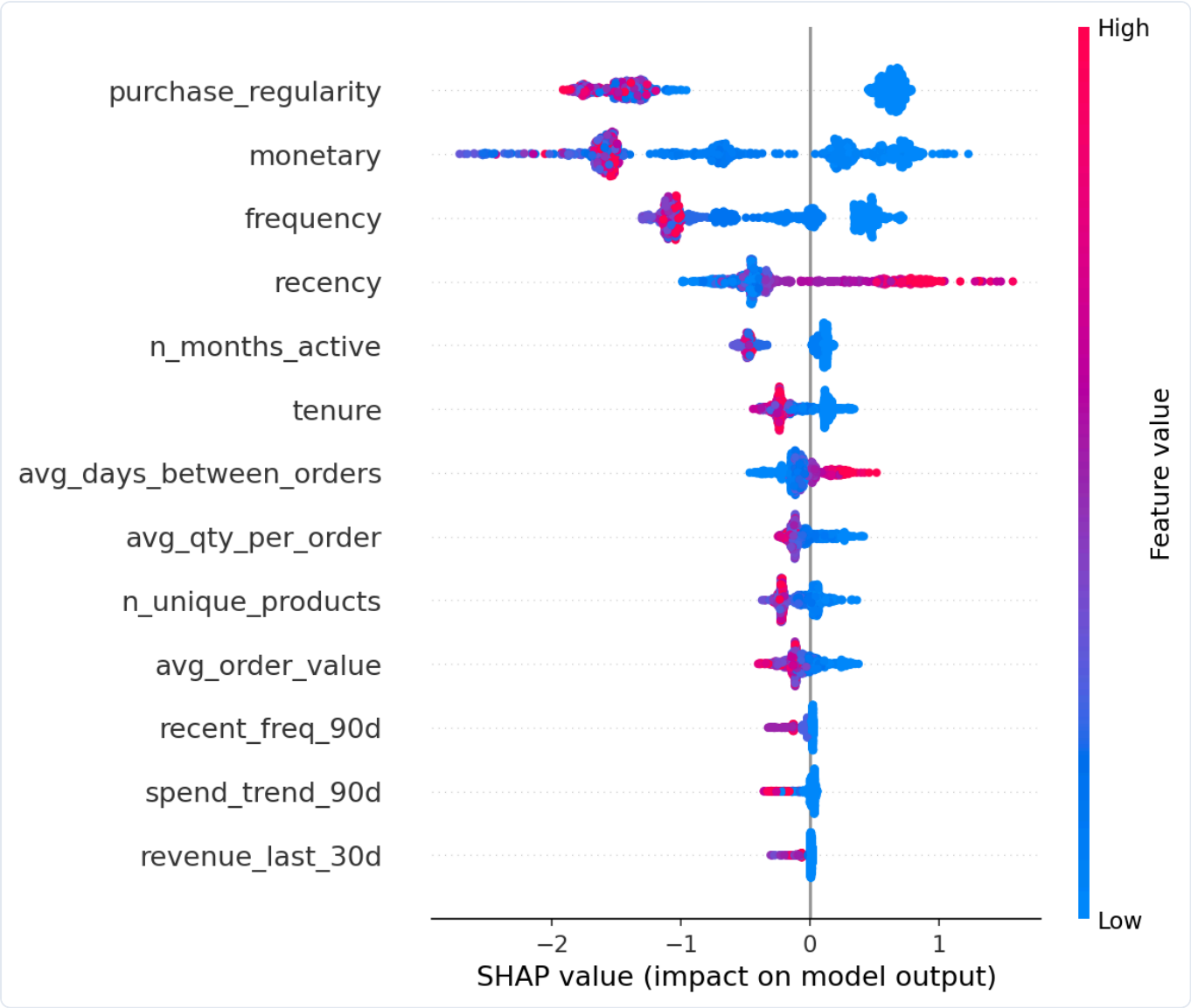
Hybrid forecast vs actuals



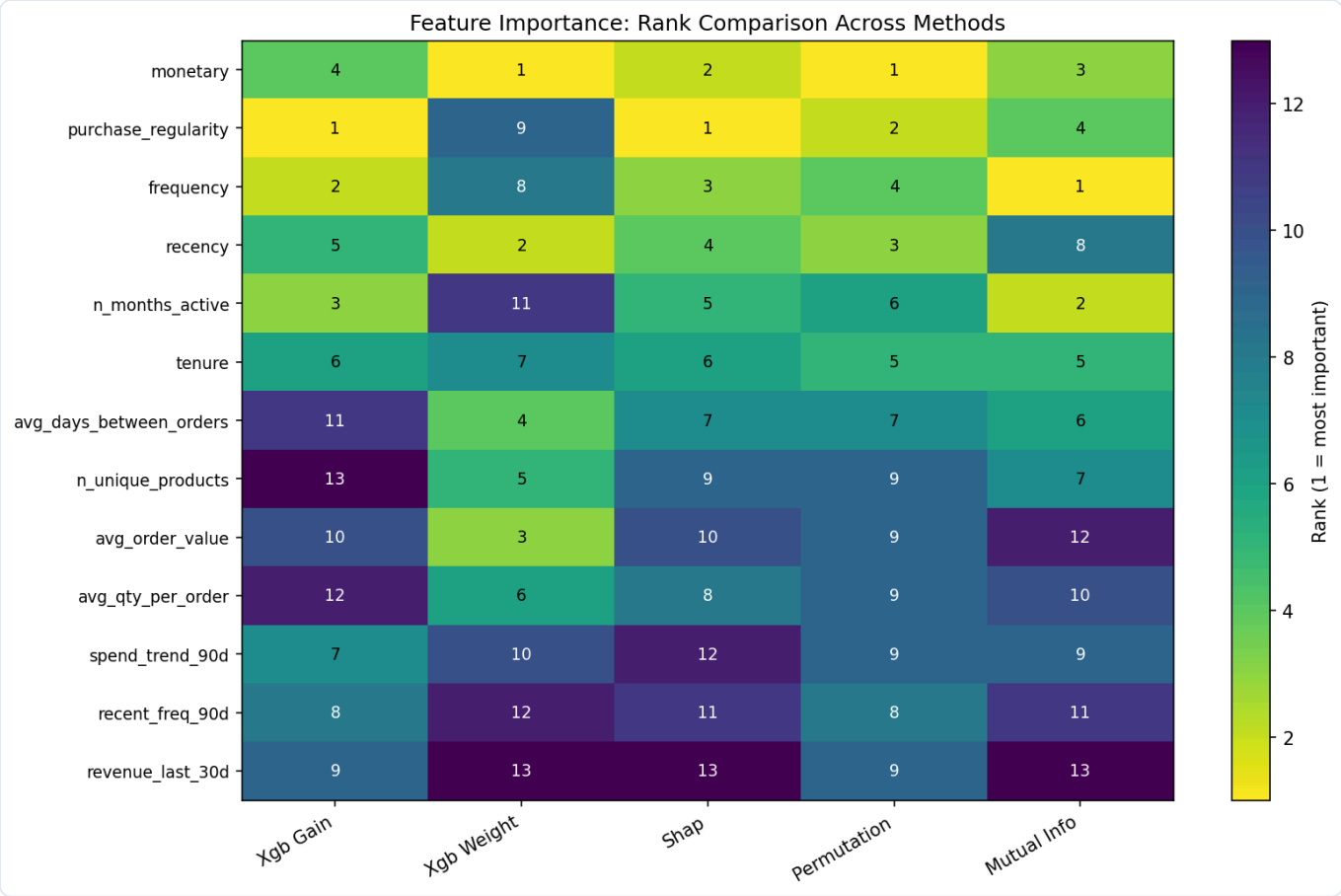
Churn ROC curve



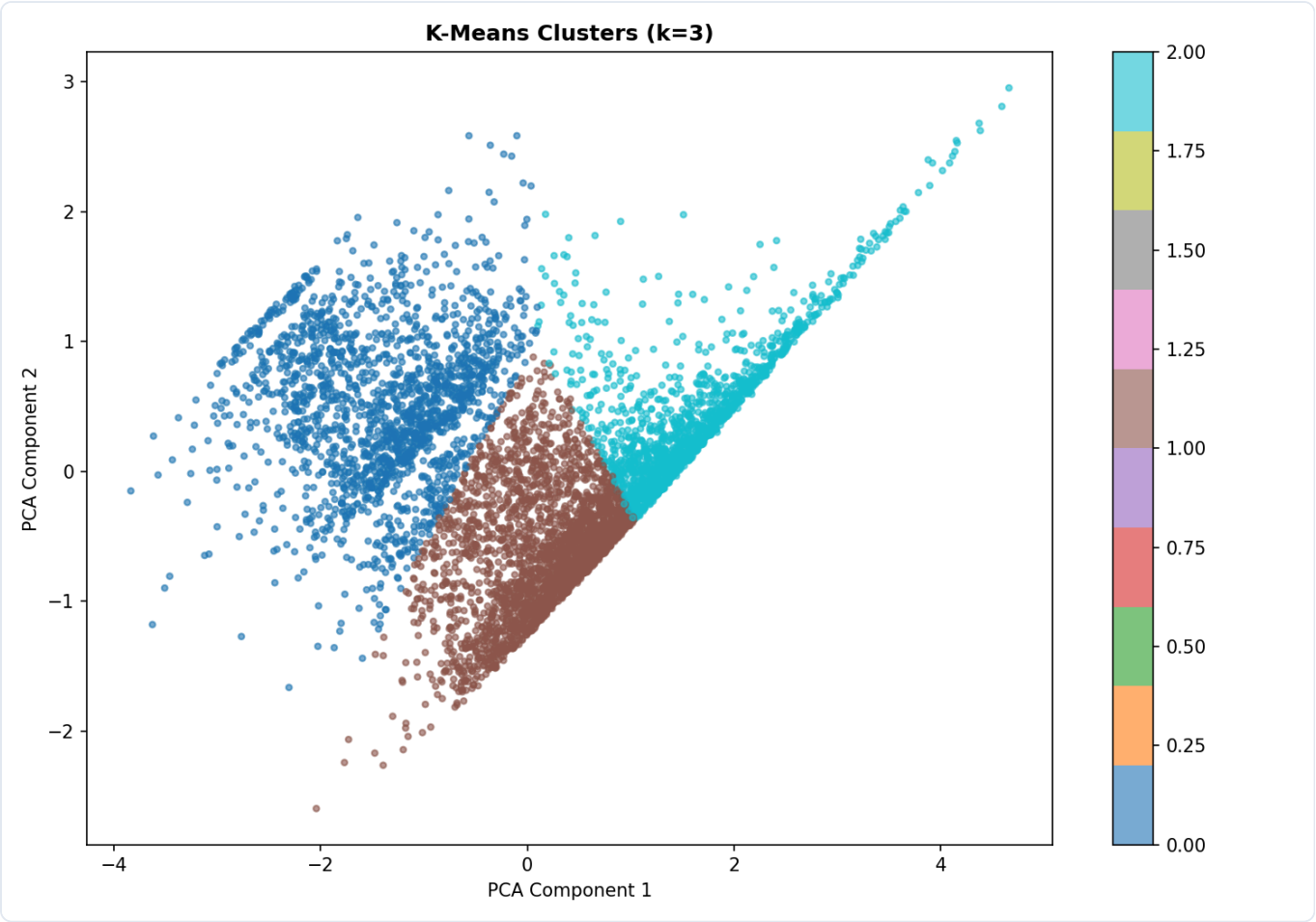
Churn SHAP summary



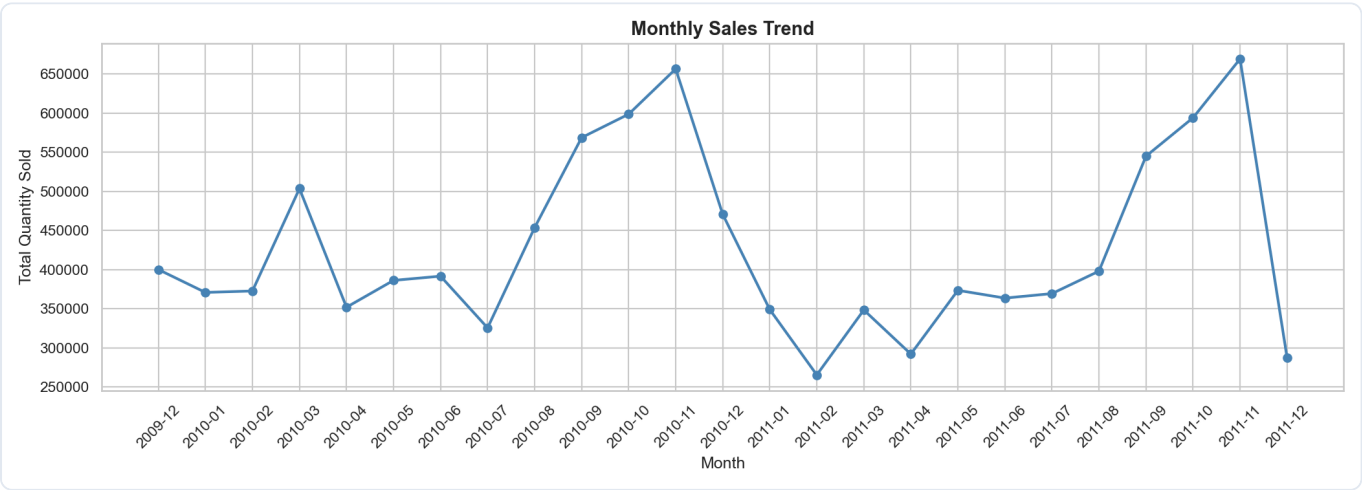
Feature importance consensus



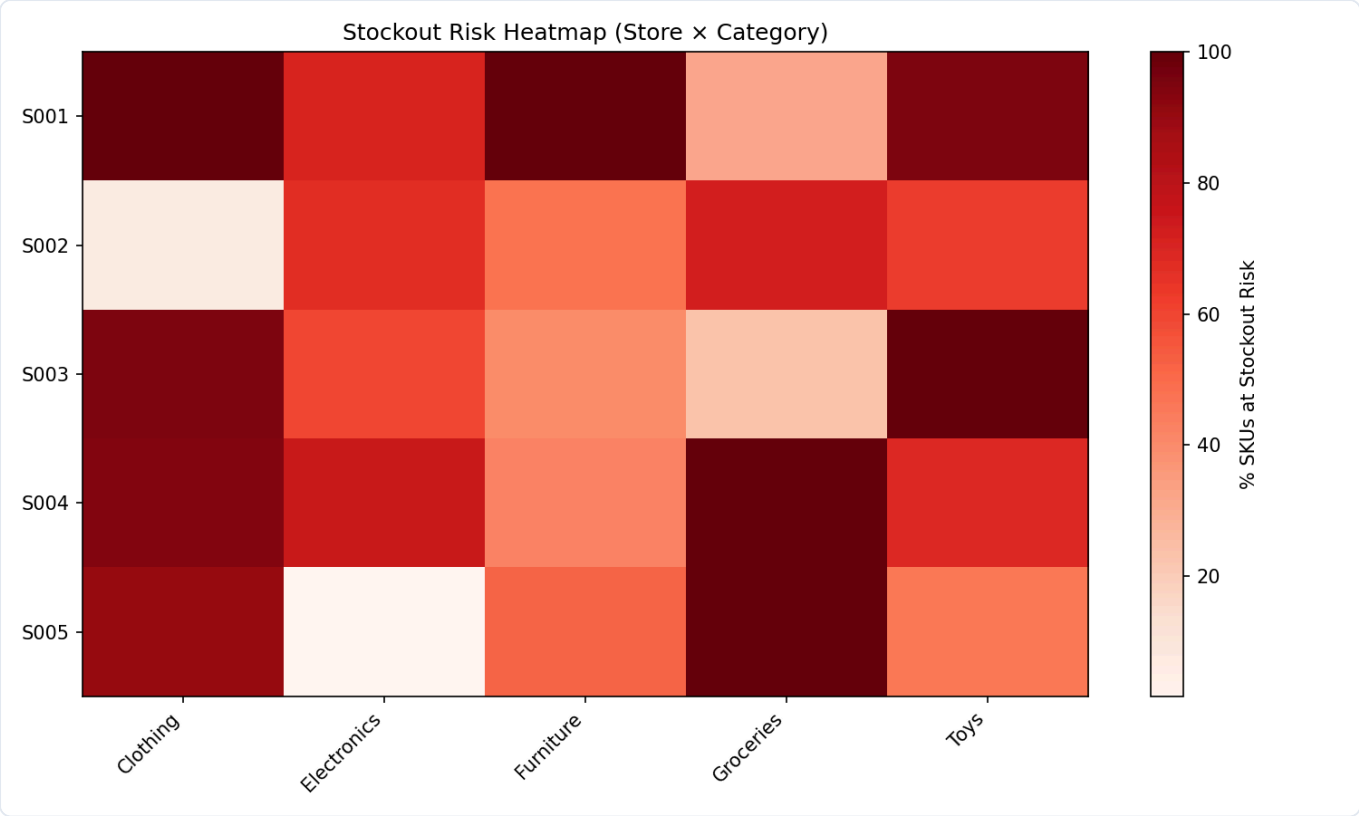
Customer segments (K-Means)



Monthly sales trend (EDA)



Inventory stockout risk heatmap



9. Personal Reflection

Key learnings.

- **Production ML is mostly engineering.** Data plumbing, fallbacks, authentication, deployment, and monitoring demanded far more effort than model training itself.
- **Hybrid models win when the signals differ.** Prophet captures seasonality while the LSTM models the residual non-linearity — together they outperform either model alone.
- **Honest metrics drive progress.** Reporting the true 31.2% forecast MAPE identifies the next optimization target far more usefully than a flattering number would.

Industry best practices applied.

- **Release-gated tracking** — MLflow baselines block any accuracy regression from reaching production.
- **Drift-triggered retraining** — Evidently and Airflow retrain models only on detected drift, behind an accuracy gate.
- **Graceful degradation** — a tiered Redis → PostgreSQL → CSV cascade keeps the application available under partial outages.
- **Automated quality gates** — CI enforces lint and tests; CD enforces an accuracy gate before every deploy.
- **Security by default** — role-based access control and strict secret hygiene throughout.

Future roadmap.

- **Improve forecast accuracy** — introduce exogenous regressors (promotions, holidays, weather) and per-category models.
- **Move to streaming ingestion** — replace batch CSV loads with continuous online data.
- **Add model-performance monitoring** — alert on prediction quality, not infrastructure metrics alone.
- **Measure intervention impact** — A/B test churn-prevention actions to quantify revenue retained.

RetailPulse — built as a machine learning internship project at Zidio Development, June 2026.